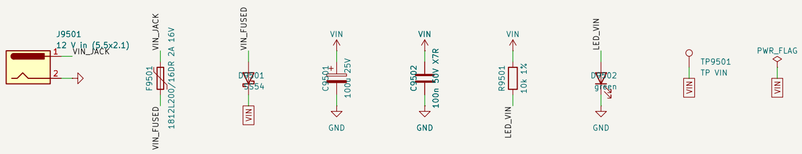
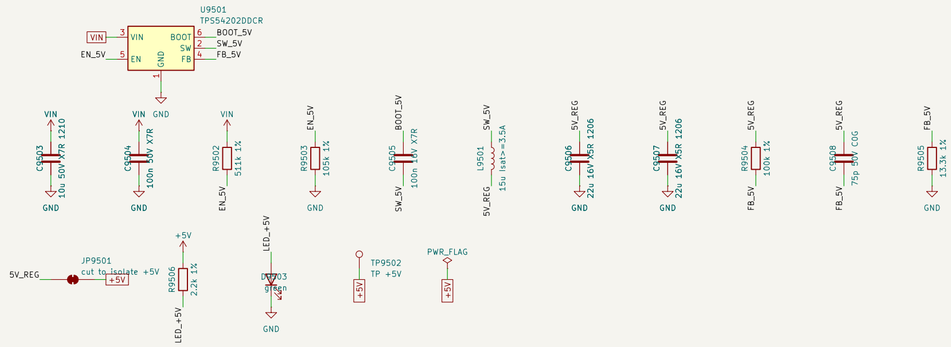
A. 12 V INPUT: jack, polyfuse (16 V), SS54 reverse-polarity diode -> VIN

Use a regulated 12 V adapter only: LM2731 VIN max 14 V. Jack 5.5 x 2.1 mm, >= 1.5 A.



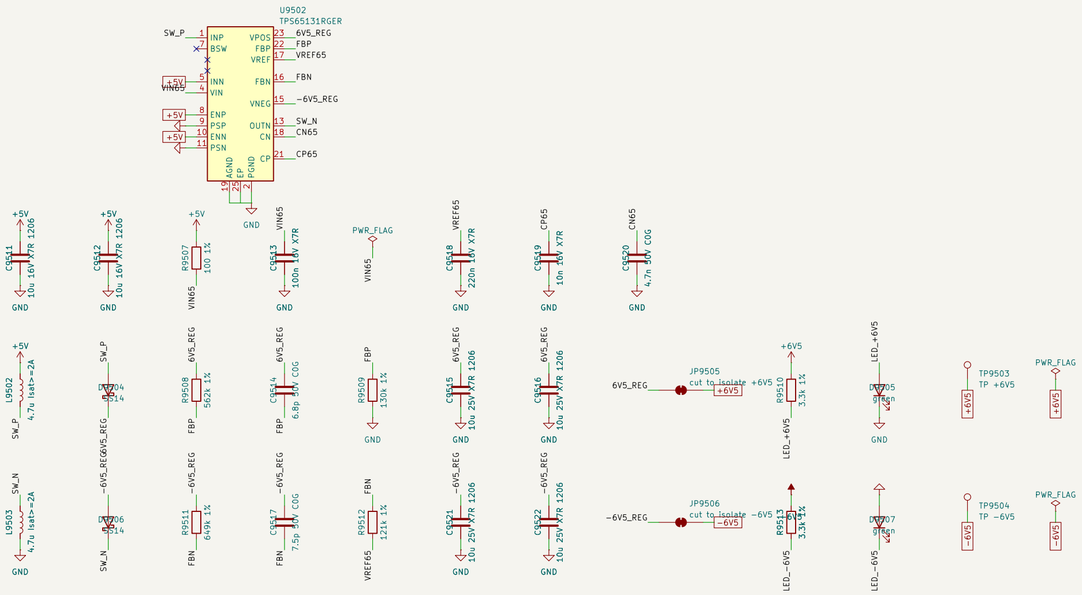
B. +5V: TPS54202 buck (TI reference design), 2 A

Vout = 0.596 V x (1 + 100k/13.3k) = 5.08 V. EN UVLO about 7 V. Load about 1.4 A.



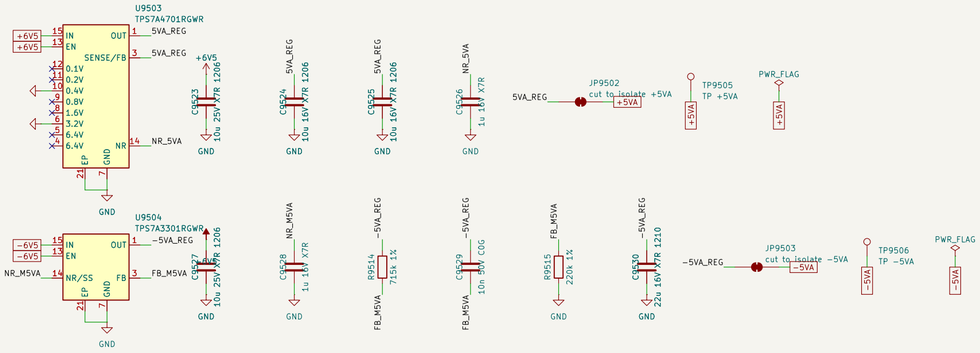
C. +-6V5: TPS65131RGER boost + inverter from +5V (feeds the analog LDOs only)

Drawn with the pin-identical TPS65130 symbol: order TPS65131RGER (2 A switches). PSP/PSN low: fixed-frequency PWM.
+6V5 = 1.213 V x (1 + 562k/130k) = 6.46 V. -6V5 = -1.213 V x 649k/121k = -6.51 V.



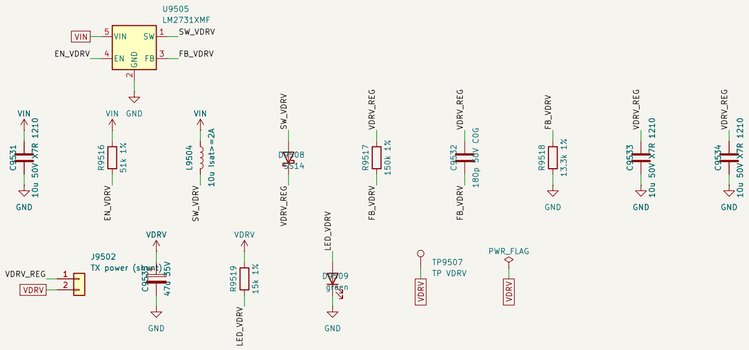
D. ANALOG RAILS: TPS7A4701 +5VA, TPS7A3301 -5VA (low noise, no LEDs)

+5VA: ANY-OUT 1.4 + 3.2 + 0.4 = 5.0 V. -5VA = -1.176 V x (1 + 715k/220k) = -5.00 V.
Load: 64 x OPA1678 (4 mA per channel) + 8 x AD7606C AVCC: about 0.45 A on +5VA, 0.3 A on -5VA.



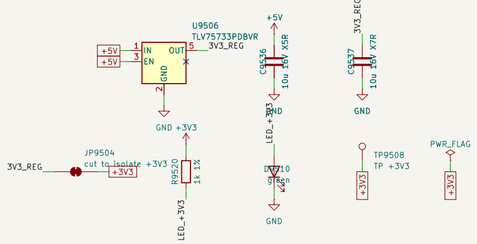
E. VDRV 15 V: LM2731X boost from VIN. Remove the J2 shunt to unpower all TX drivers

1.23 V x (1 + 150k/13.3k) = 15.1 V. Load: 64 elements CW at 40 kHz, CV^2f = 1.3 W, about 0.12 A.



F. +3V3: TLV75733P LDO from +5V (1 A, stable with ceramic caps)

Load < 0.15 A: ADC VDRIVE, clock buffers, logic, mics.

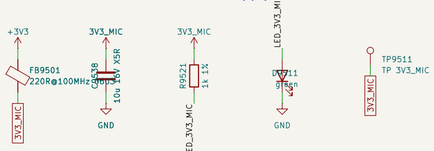


G. GROUND: one GND net and plane

Test points for scope grounds.



H. 3V3_MIC: ferrite-filtered supply for the 8 MEMS mics (remove FB1 to isolate)



Hybrid 8x8 ultrasonic AESA
Sheet: /Power/
File: power.kicad_sch
Title: Power: 12 V in, +5V, +-6V5, +-5VA, VDRV, +3V3, 3V3_MIC
Size: A2 | Date: 2026-10-20 | Rev: 2
KiCad E.D.A. 10.0.6 | Id: 115/115